

Combining DAS Ambient Seismic Noise Interferometry and Unsupervised Machine Learning for 2D Imaging of an Urban Subsurface

Leila Ehsaninezhad^{*1,2}, Christopher Wollin¹, Verónica Rodríguez Tribaldos¹, and Charlotte Krawczyk^{1,2}

Abstract

Applying ambient noise interferometry to distributed acoustic sensing (DAS) data recorded along telecommunication networks offers a promising way to image the urban subsurface with high resolution at reduced costs. However, extracting coherent seismic signals from DAS in urban environments can be challenging due to diverse, unevenly distributed noise sources, which can distort interferometry results and produce spurious signals, complicating analysis and interpretation. To address these complexities, we have developed a modified ambient noise interferometry workflow that allows for the efficient selection of high-quality data. We analyze 15 days of continuous passive DAS data collected from a pre-existing 11 km long dark fiber-optic cable running along a major urban road in Berlin, Germany. After retrieval of virtual shot gathers (VSGs) using the cross-correlation method, we develop a selection strategy to identify high-quality data through unsupervised clustering. Next, we stack the VSGs within each cluster and choose the highest quality, stacked VSG based on the quality of the corresponding dispersion spectra. We initially test the clustering method on synthetic VSGs to ensure its effectiveness. The clustering results identify distinct groups of VSGs that exhibit consistent patterns in both synthetic and real VSGs. These distinct groupings offer valuable insights into the temporal variations in human activities and allow a better interpretation and identification of viable ambient noise signals for further processing. Thereafter, multichannel analysis of surface waves is utilized to obtain 1D shear-wave velocity models for consecutive array segments. A 2D subsurface velocity model is then constructed through merging the individual 1D velocity models derived from overlapping array subsections. This approach enables efficiently selecting highest-quality data within massive, noisy recordings, which ultimately results in enhanced dispersion measurements and thus improved images of the urban subsurface.

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[Supplemental Material](#)

Introduction

Ambient seismic noise interferometry is a powerful approach for various geophysical applications, including imaging and seismic monitoring of subsurface structures (Campillo and Roux, 2015; Lehujeur *et al.*, 2016). Unlike traditional seismic methods that rely on active sources, ambient noise interferometry utilizes the ambient seismic noise generated by natural sources and anthropogenic activities. This makes it ideal for urban environments in which employment of active seismic sources is costly and human-generated seismic noise is widespread and abundant. The technique of ambient noise interferometry is commonly based on the cross correlation of ambient seismic noise recordings between two locations

(Campillo and Roux, 2015; Lehujeur *et al.*, 2016). By cross-correlating ambient noise records collected over a period of time, such as minutes, hours, or even days, we can obtain the noise cross-correlation function (NCF). Analysis of the obtained NCF yields valuable information about the seismic velocity

1. Helmholtz Centre for Geosciences (GFZ), Potsdam, Germany, <https://orcid.org/0000-0002-0283-4557> (LE); <https://orcid.org/0000-0002-3992-787X> (CW); <https://orcid.org/0000-0002-1325-0434> (VRT); <https://orcid.org/0000-0002-5505-6293> (CK); 2. Institute of Applied Geosciences, Technical University of Berlin, Berlin, Germany

*Corresponding author: leila@gfz.de

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and thus the Earth's subsurface structure between the two locations of receivers (Bensen *et al.*, 2007; Boué *et al.*, 2014).

Anthropogenic activity creates ambient noise with a frequency content particularly favorable for imaging the shallow subsurface (Nakata *et al.*, 2011; Song *et al.*, 2021; Tribaldos *et al.*, 2021). However, the presence of complex noise sources can affect the quality of the extracted NCF and thus poses a key challenge for effective ambient noise interferometry. Spurious signals in the NCF may arise from the subsurface structure (Yang, Zhan, *et al.*, 2022) or the uneven distribution of noise sources (Lehujeur *et al.*, 2016; Girard and Shragge, 2020). The latter can lead to biases in the measured seismic velocities. As a result, careful quality control of the NCF or denoising techniques is crucial for improving the reliability and accuracy of ambient noise interferometry results (Girard and Shragge, 2020). To address this issue, researchers have developed various techniques, including denoising approaches such as discrete orthogonal S-transform (Baig *et al.*, 2009), singular value decomposition (Moreau *et al.*, 2017), frequency-wave-number filtering (Li *et al.*, 2023), coherence-based wavefield decomposition (Ehsaninezhad *et al.*, 2024), isolating the wavefield generated by a specific noise source, such as vehicles (Yuan *et al.*, 2024) or ensuring quality control of the recorded ambient noise (Chang and Nakata, 2024).

Stacking consists of the superposition of NCFs obtained at different time periods and aims at improving the quality of the NCF by enhancing the SNR and preserving phase and amplitude information (Yang *et al.*, 2023). Some commonly used stacking methods include linear stacking (Bensen *et al.*, 2007) and phase-weighted stacking (Schimmel and Paulssen, 1997; Cheng *et al.*, 2021). Some studies conduct a selection of the suitable data before applying either of these stacking approaches, what is referred to as selective stacking (Safarkhani and Shirzad, 2019; Xie *et al.*, 2020; Thangraj and Pulliam, 2021; Pedersen *et al.*, 2023; Yang *et al.*, 2023). The choice of stacking method depends on the specific research application, for example, seismic tomography, monitoring of velocity changes, or characterization of seismic attenuation properties (Yang *et al.*, 2023). Generally, for tomography purposes, all stacking methods yield comparable results (Yang *et al.*, 2023). However, phase weighted stacking is a nonlinear operation and suffers from waveform distortion especially when incoherent signals are present (Cheng *et al.*, 2021).

Selective stacking can enhance the quality of the retrieved signals (Xie *et al.*, 2020; Thangraj and Pulliam, 2021; Pedersen *et al.*, 2023) and is based on a detailed analysis of the NCFs focusing on their spatial and temporal variations. This process typically relies on various metrics, including the root mean square (rms) ratio of signal and noise windows (Xie *et al.*, 2020), similarity criteria (Yang *et al.*, 2023), and the amplitude ratio between horizontal and vertical components (Pedersen *et al.*, 2023). The similarity criterion is usually applied between each individual NCF and a reference NCF, which is often the

median or mean of all NCFs. This approach can introduce bias, especially when there is a significant number of low-quality NCFs. An alternative method is to group the NCFs into clusters (Viens and Iwata, 2020; Yates *et al.*, 2023; Fang, 2024). Viens and Iwata (2020) proposed the idea of clustering the NCFs using a dimensionality reduction technique (principal component analysis, PCA). They successfully distinguished and removed noisy NCFs from high-quality ones by clustering the pairwise deconvolution of NCF time series.

Recently, the combination of ambient noise interferometry and distributed acoustic sensing (DAS) has been successfully applied in various settings, including subsurface imaging (Ajo-Franklin *et al.*, 2019; Shragge *et al.*, 2021; Tribaldos *et al.*, 2021; Yang, Atterholt, *et al.*, 2022; Cheng *et al.*, 2023) and monitoring (Rodríguez Tribaldos and Ajo-Franklin, 2021; Maass *et al.*, 2024; Shen *et al.*, 2024). The fundamental principle behind DAS is the use of fiber-optic cables as a continuous array of sensors capable of detecting and measuring strain imposed on any segment along the entire length of the cable (Hartog, 2017; Lindsey *et al.*, 2020). This allows for seismic arrays with a very high spatial resolution which would be achievable with traditional point sensors only at a high cost. By leveraging DAS deployed on pre-existing, unused telecommunication cable (dark fiber) and seismic noise interferometry, we can obtain NCFs between each pair of channels. If we divide the cable into segments, the concatenation of individual NCFs with a single reference source at various receiver locations within each segment produces a virtual shot gather (VSG). In this way, besides the temporal dimension the spatial dimension is additionally resolved.

In this article, we aim to improve the ambient noise interferometry workflow by assessing the quality of VSGs derived from complex urban ambient noise data through unsupervised clustering. Our objective is to enable efficient high-resolution 2D seismic subsurface imaging using 15 days of continuous passive DAS data recorded on dark fiber in an urban area of Berlin, Germany. An additional goal is to investigate the effects of temporal variations in human activities in the identification of viable ambient noise signals that can be further processed for imaging purposes.

We begin the article by describing the 15 days of continuous DAS urban noise data and the interferometric process resulting in a large set of VSGs associated to different recording times and fiber segments. We then follow Viens and Iwata (2020), who proposed a clustering method for NCFs (1D) obtained from seismic noise data of sparse pairs of seismometers. We extend their approach to the 2D VSGs obtained from DAS urban noise by converting the VSGs to wavelet coefficients prior to the clustering. In the following, we first validate the extended clustering method with a synthetic dataset and then apply the technique to the VSGs obtained from overlapping array subsections of the cable. Afterward, we present the VSG stacks for each cluster and their dispersion spectra as well

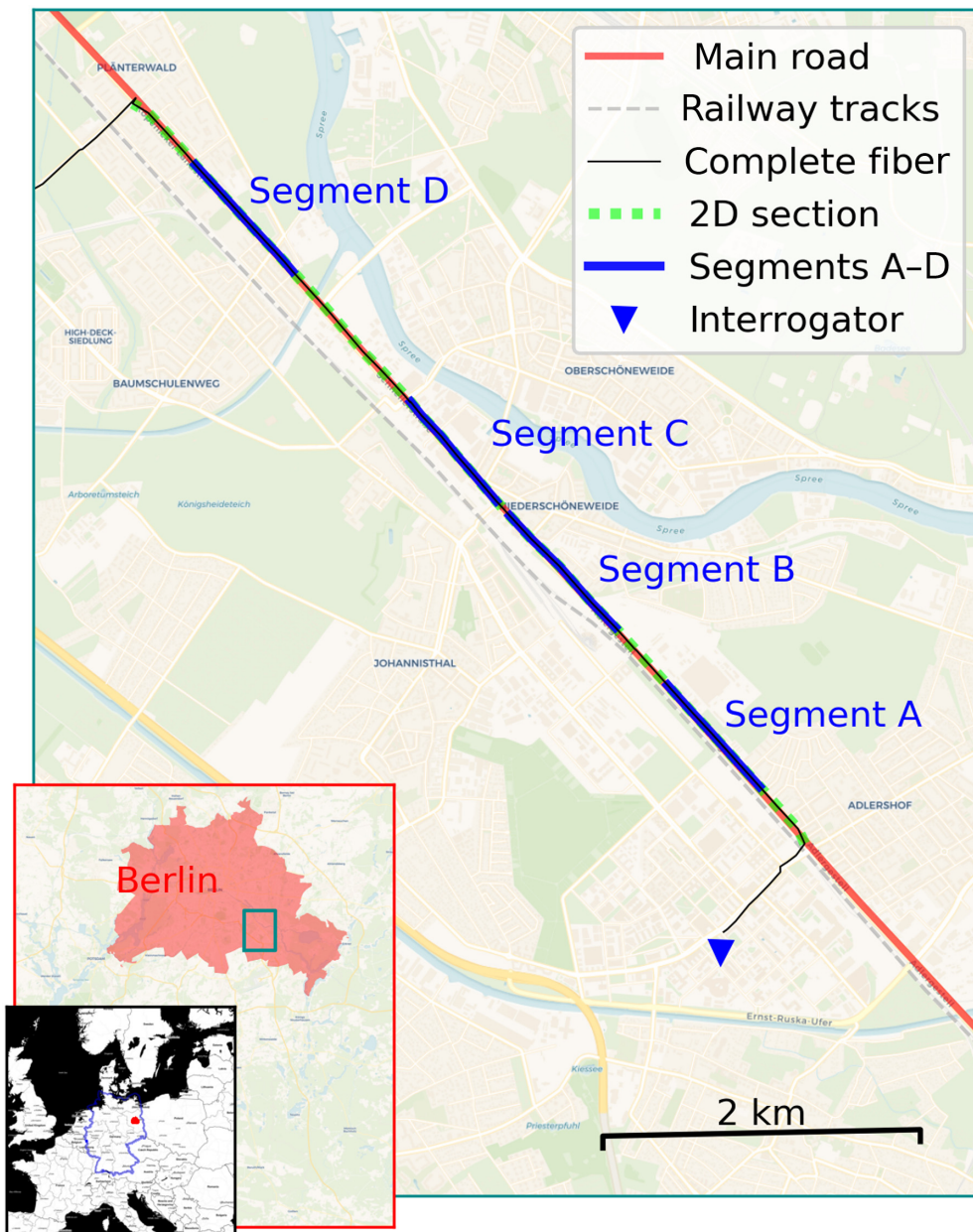


Figure 1. Map of the study area (OpenStreetMap contributors, 2017, data by OpenStreetMap). The insets in the lower left depict Adlershof's location (cyan frame) within Berlin in Germany and central Europe (OpenStreetMap contributors, 2017, data by OpenStreetMap, map tiles by Stamen Design). The plot shows the interrogator's position (blue triangle), the fiber trajectory along the major road "Adlergestell" (red), a parallel railway track (gray dashed line), and the 2D section location (green dashed line). For example, 1 km long segments of cable used to illustrate the methods and results of this article are highlighted in blue. The color version of this figure is available only in the electronic edition.

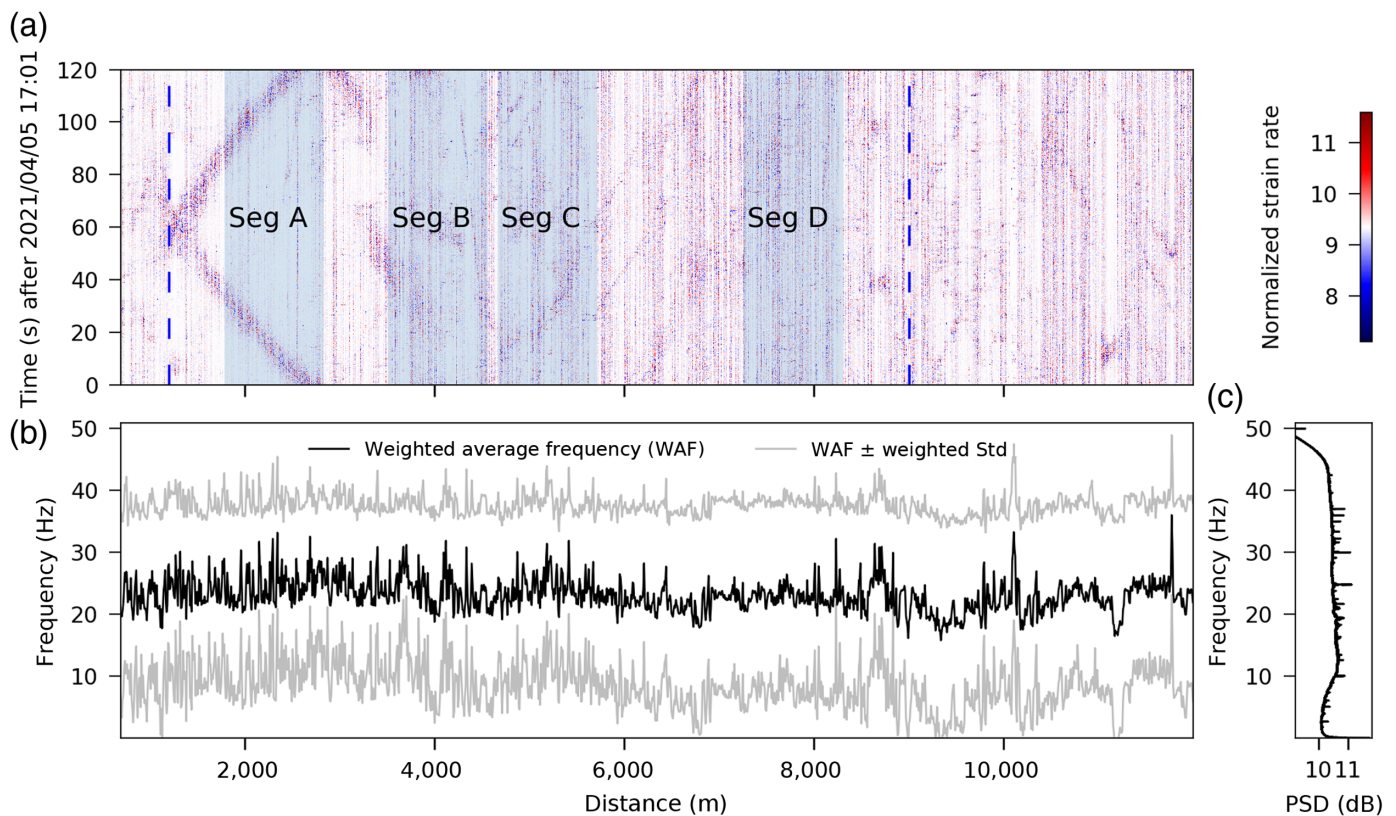
as the result from the multichannel analysis of surface waves (MASW) applied on the best VSG stack of each cable segment, along with the final 2D shear-wave velocity model of the subsurface and an analysis on DAS data usability (hours of the day and night). We conclude the article with a discussion on assessing the effectiveness of clustering and validating the constructed 2D velocity model.

Study Site and Data Description

The area of study is the Adlershof neighborhood in the southeast of the German capital, Berlin (Fig. 1). Geologically being part of the North German basin, the area is known for its geothermal potential. This basin is a large sedimentary basin with high potential for geothermal energy. This is largely because of thick Mesozoic sandstone layers, which were formed over millions of years and have good water flow properties, making them ideal for extracting under-ground heat (Frick *et al.*, 2022; Norden *et al.*, 2023). As a consequence, Berlin hosts numerous projects related to geothermal energy extraction or high-temperature Aquifer Thermal Energy Storage (ATES), for which investigating the subsurface structure is critical to ensure safe and efficient utilization. Of particular interest is a subsurface clay layer known as the Rupelian clay, which acts as a hydraulic barrier between freshwater and saline aquifers. The latter is the potential aquifer for ATES (Pei *et al.*, 2024) and thus must be carefully studied to avoid environmental hazards derived from geothermal energy production, such as groundwater salinization. Accurate delineation of this layer is essential for safe geothermal energy development.

To investigate the subsurface in this region, we utilized a straight 8-km-long section of a

dark telecommunication fiber as an array of sensors to measure urban noise (e.g., traffic) along a major road. Strain-rate measurements were conducted with a commercial Silixa iDAS unit at a spatial resolution of 8 m and a temporal frequency of 1 kHz with a gauge length of 10 m. Further details regarding the recording, the fiber, and the geology of the site are given in Ehsaninezhad *et al.* (2024).



The channel-normalized strain rate for a 2 min DAS record along the 11-km-long fiber is illustrated in Figure 2a in which the seismic features correspond to seismic noise primarily generated by moving vehicles such as buses and trains traveling parallel to the fiber-optic cable. The most prominent events are zig-zag patterns created by trains moving in opposite directions. Figure 2b shows the amplitude-weighted average frequency and the respective standard deviation across time of the power spectral density for two-day-long time series of each channel. In addition, the average power spectral density (PSD) over all channels is shown in Figure 2c. These figures indicate that the dominant spectral energy lies between 8 and 40 Hz (Fig. 2b). More details about the frequency content of the data over the 15 days of recording are described in Ehsaninezhad *et al.* (2024).

Method

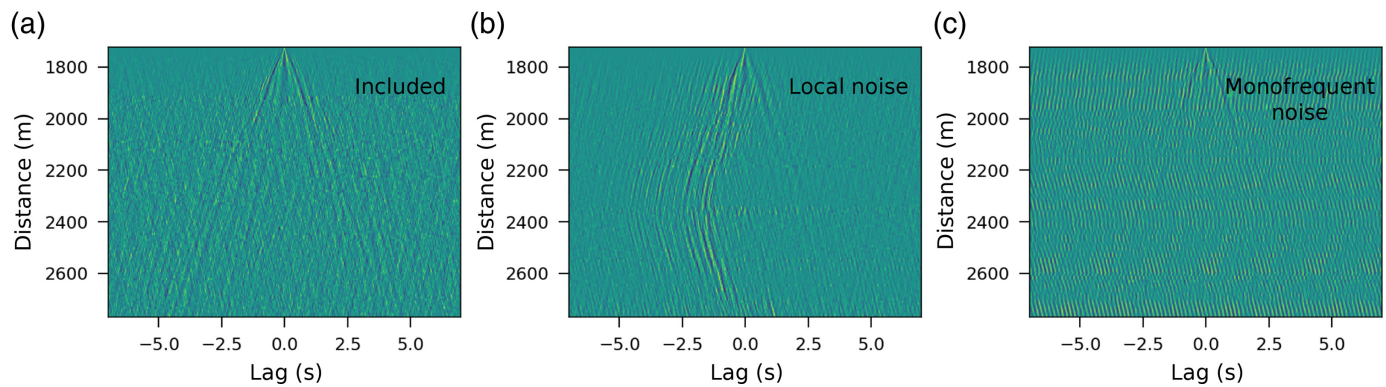
Interferometric processing of DAS ambient noise

The 2D subsurface shear-wave velocity model we aim for is obtained by concatenating a set of stringed 1D velocity profiles. The cable is segmented using a rolling window approach with 90% overlap along the entire straight section of the fiber, whereas channels laying within loops of cable slack are removed (also see Ehsaninezhad *et al.*, 2024). For each segment, we follow the ambient noise data preprocessing workflow presented in Bensen *et al.* (2007) and calculate cross correlations for each hour-long DAS data time series of the southeastmost channel with its neighboring channels in each 1 km segment of the cable,

Figure 2. (a) Color-coded strain-rate amplitude of a 2 min long DAS recording. The zig-zag lines indicate the imprints of trains traveling along the cable. The blue shaded areas represent segments A–D (see supplemental material for details on segments B, C, and D) discussed in this article. The blue lines indicate the locations of bends in the cable and frame the straight segment analyzed in this article (green dashed in Fig. 1). (b) Amplitude-weighted average frequency (WAF in black) and the respective standard deviation (in gray) across time for two-day-long distributed acoustic sensing (DAS) recordings calculated for each channel. (c) The spatially averaged power spectral density (PSD) over all channels, where the signal level is given concerning the resolution of the measurement, that is, $\approx 1.4 \text{ nmm}^{-1} \text{ s}^{-1}$. The color version of this figure is available only in the electronic edition.

similar to our workflow in Ehsaninezhad *et al.* (2024). With this approach, we generate VSGs for each hour of the day over the entire recording period totaling 360 VSGs for each of the 1-km-long fiber segment.

VSGs are frequently contaminated by strong local sources of noise. Figure 3 shows three exemplary patterns identified in segment A (also compare Figs. 1 and 2). Before stacking, VSGs with spurious arrivals and high noise levels need to be identified and excluded to achieve the largest possible improvement of the SNR. Figure 3a shows a good-quality VSG, with a symmetric pattern of surface waves moving across the array, whereas Figure 3b,c shows low-quality gathers, displaying a localized noise source deviating the surface-wave moveout, and



monofrequent noise contaminating the entire VSG, respectively. After identifying and excluding VSGs of the latter two types, the suitable VSGs are linearly stacked.

This segmented interferometric processing of DAS ambient noise divides the long, straight segment of the cable shown in Figure 1 into 80 overlapping 1-km-long subsegments with a virtual source spacing of 96 m. Visual inspection and manual selection of VSGs as conducted by Ehsaninezhad *et al.* (2024) for a single segment (segment A) may be feasible for the respective 1D analysis. However, the expansion from 1D to 2D analysis is too time consuming with this procedure, which highlights the need for an automated method to identify and remove low-quality VSGs. To achieve this, we employ unsupervised clustering (Viens and Iwata, 2020), a machine learning method for dividing a dataset into groups with similar characteristics (Steinbach *et al.*, 2004).

Clustering of VSGs

In the context of a set of NCFs obtained from a specific pair of stations but different time intervals, Viens and Iwata (2020) proposed a clustering approach based on features derived from a reduced representation of the NCFs (time series of NCFs). Their two-step approach consists of first applying PCA (Abdi and Williams, 2010) to reduce the dimensionality of the set of NCFs. Then, a Gaussian Mixture Model (GMM) (Figueiredo and Jain, 2002; Viens and Iwata, 2020) is used to cluster the extracted principal components.

PCA is a statistical method for dimensionality reduction that extracts key information from data and represents it using new orthogonal variables called principal components (Abdi and Williams, 2010). These components are ranked by the variance they explain, with the first component capturing most of the variance. For a detailed explanation of PCA, we refer the reader to Abdi and Williams (2010).

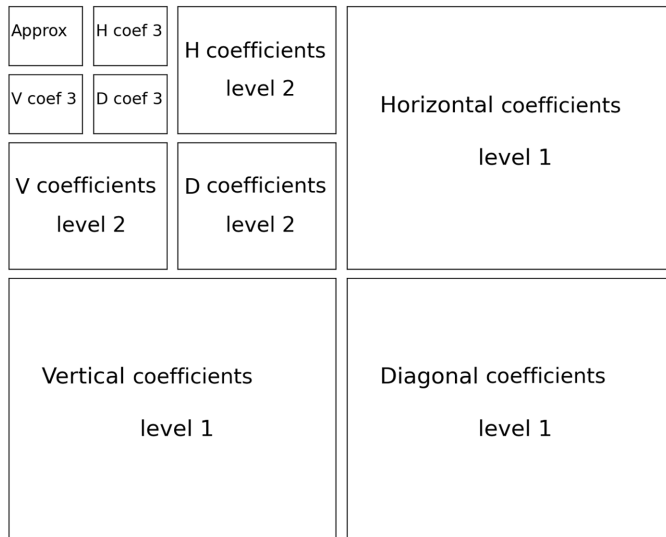
A GMM is a probabilistic clustering method that models data as a mix of Gaussian distributions with unknown parameters (Figueiredo and Jain, 2002). It extends k-means by considering data covariance and assigns points to clusters based on likelihood (Pedregosa *et al.*, 2011). For a more detailed utilization of PCA and GMM refer to Viens and Iwata (2020).

Figure 3. Comparison of three different types of virtual shot gathers (VSGs) resulting from cross-correlating data recorded at different times throughout the day (a) included gather and (b) excluded gather, showing artifacts caused by the presence of a strong local noise source. (c) Excluded from gathering with monofrequent noise pattern. (Note that the distances shown in the figures represent the distances of the channels from the interrogator unit.) The color version of this figure is available only in the electronic edition.

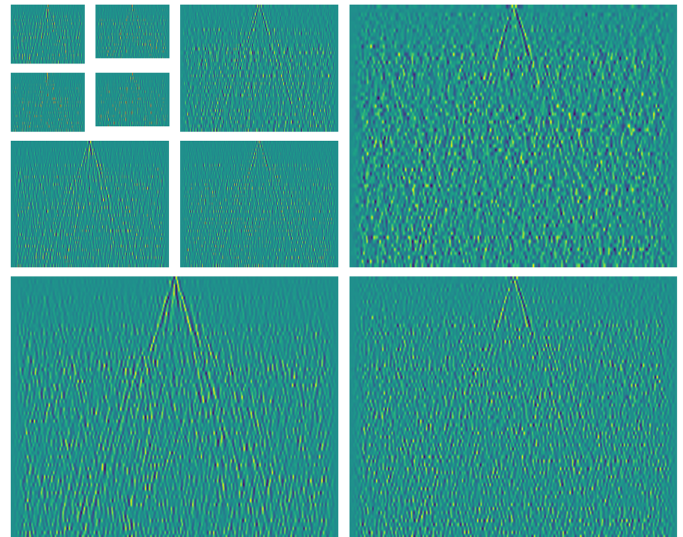
Extracting features that capture decisive information from each sample (in our case, VSGs) is essential for effective clustering. Whereas Viens and Iwata (2020) work with sets of 1D time series (NCFs), our study considers sets of VSGs, which resolve both time and space and are therefore 2D. Similar to their approach, we cluster all VSGs corresponding to one fiber segment to identify different VSG patterns and discard those of low quality. However, we modify the clustering method to better suit the 2D structure of the VSGs by adding an additional step prior to data reduction. Instead of applying PCA directly to the VSGs, we first transform the spatiotemporal data into wavelet coefficients using the discrete wavelet transform (DWT) as described in the next section. Compared to the principal components obtained from direct PCA of the set of VSGs, or other representations like PSD, the PCA of the wavelet coefficients yields a larger variance (Fig. S6, available in the supplemental material to this article) and therefore a better separation of the VSG patterns. We then reduce the dimensionality of the obtained wavelet coefficients by keeping the most decisive principal components before finally clustering the reduced data. Furthermore, following Ermert *et al.* (2023) we predefine the number of clusters to two or three based on the noise patterns as observed in this urban environment in the course of a day (Ehsaninezhad *et al.*, 2024). The optimal number of clusters (two or three) is finally determined during an initial clustering trial.

DWT. The VSG waveforms are initially subjected to the DWT to derive the wavelet coefficients (Lee *et al.*, 2019). DWT is founded on the concept of multiresolution analysis introduced by Mallat (1989), Anant and Dowla (1997), and Oonincx

(a)



(b)



(1999). It decomposes a signal into a set of basis functions that reveal detailed information and patterns within the data (Mallat, 1989). This approach also provides enhanced capabilities for analyzing and interpreting seismic data by utilizing multiresolution analysis (Mallat, 1989; Anant and Dowla, 1997; Oonincx, 1999). Such capabilities are particularly useful in seismic data processing in which feature extraction and identification of transient events are crucial. This technique is widely applied in seismology for tasks such as denoising (Quinones and Tibi, 2024) and event detection (Anant and Dowla, 1997).

DWT encompasses an iterative filtering and down-sampling process. At each level of the wavelet transform, the input signal undergoes filtering with both low-pass and high-pass filters, resulting in two components: approximation and details. The filtered signals are then down-sampled by a factor of two, generating the DWT coefficients (Zhang *et al.*, 2003). The wavelet coefficients capture the signal's characteristics across various scales (or resolutions). Details correspond to the high-frequency elements, whereas the approximation pertains to the low-frequency elements of the signal. This enables the analysis of the signal at different resolutions (Mallat, 1989; Zhang *et al.*, 2003).

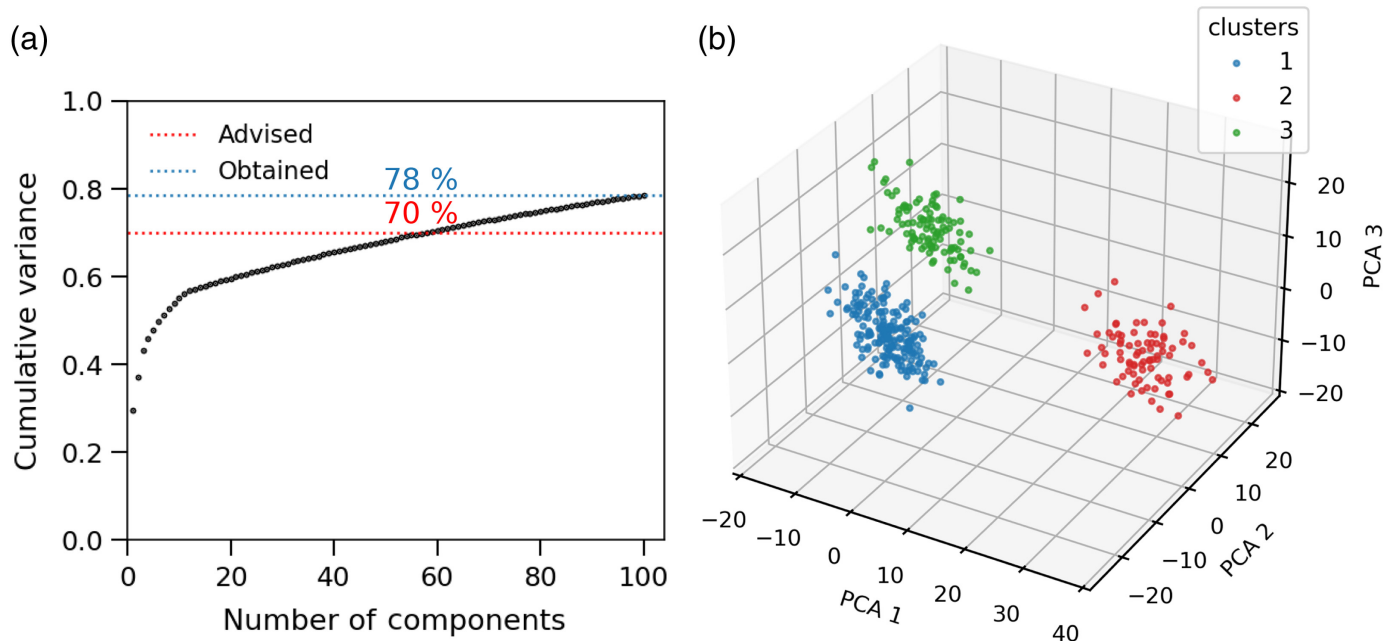
The 2D wavelet transform extends this concept to 2D signals, such as images or spatiotemporal data, allowing their decomposition into various scales and orientations. This decomposition provides localized information in both spatial and temporal domains (Anant and Dowla, 1997; Gendron *et al.*, 2000; Zhang *et al.*, 2003). Following this approach, we iteratively compute the wavelet coefficients of each VSG for three decomposition levels across the vertical, horizontal, and diagonal directions (Lee *et al.*, 2019). Figure 4 illustrates an example of wavelet coefficients for the VSG shown in Figure 3a. Note that this representation of the data contains approximately the same total number of samples across both time and space as the original dataset (also see Fig. S4).

Figure 4. An example of wavelet coefficients across three levels of decomposition: (a) Schematic illustration of the wavelet coefficients. (b) Corresponding coefficients of a VSG. The color version of this figure is available only in the electronic edition.

The wavelet coefficient values are averaged separately for each directional DWT coefficient: the horizontal coefficients across time and vertical and diagonal ones across space, for all levels. This results in 10 vectors, each representing one direction of the DWT coefficient. These averaged 1D vectors are then concatenated into a single vector containing 1052 samples (also see Fig. S4) which is then standardized by subtracting the mean and dividing by the standard deviation. The resulting characteristic vector represents one VSG and serves as a single sample in terms of the PCA and the clustering.

Workflow development on synthetic VSGs

To ensure the reliability of the method, we first developed and tested the workflow on synthetic VSGs. Following the approach outlined by Viens and Iwata (2020), we constructed VSGs using a chirp function with a frequency range between 1 and 30 Hz. The duration of the chirp signal increases with distance from the virtual source, within a window defined by two diverging linear functions. Besides a VSG with normal surface-wave moveouts, we also simulate two types of VSGs contaminated by noise, mimicking the noise patterns observed and shown in Figure 3. For the VSG affected by the presence of a local noise source (Fig. 3b), we deviate one of the two linear edges of the chirp function. Regarding VSGs contaminated with harmonic noise (Fig. 3c), we add several monofrequent signals to 50 randomly selected channels out of the 132 channels of the VSGs. We create 180 undisturbed VSGs, 100 VSGs polluted by local noise sources, and 80 VSGs polluted by monofrequent noise, totaling 360 synthetic VSGs, of which



one example of each is presented in Figure S1. To test the discrimination of the VSGs by our clustering approach, we additionally add noise to the simulated VSGs. The noise amplitude is randomly sampled from a Gaussian distribution with a mean of 0 and a standard deviation of 1, such that the latter is in the order of the signal amplitude.

To perform clustering, we start by randomly shuffling the 360 VSGs. From there, we calculate three levels of DWT coefficients across three directions. An example of the calculated DWT wave coefficients is presented in Figure S2. As described earlier, we then calculate the 1D representative vector of each VSG by averaging and concatenating across all levels and directions. After applying the PCA, we reduce the data dimensions retaining the first hundred principal components, which account for 78% of the cumulative variance in the data (blue dotted line in Fig. 5a). We should note that this exceeds the commonly recommended threshold of 70% (Jolliffe, 2002) (red dotted line in Fig. 5a). Clustering is then performed using a GMM with three clusters on the first 100 principal components. The PCA projection onto the first three principal components is visualized in Figure 5b. In this scenario, the three types of VSGs are clustered with 100% accuracy. Thereafter, we stack the waveforms of each cluster as shown in Figure S3.

Results

We present the results of clustering the VSGs obtained from 15 days of DAS ambient noise through the exemplary segment A indicated in Figure 1. The clustering functionality is further investigated using four independent, nonoverlapping fiber segments (segments B, C, and D), which are also highlighted in blue in Figure 1. In addition, to evaluate the data quality of each VSG cluster, we inspect the resulting VSG stack and the respective dispersion.

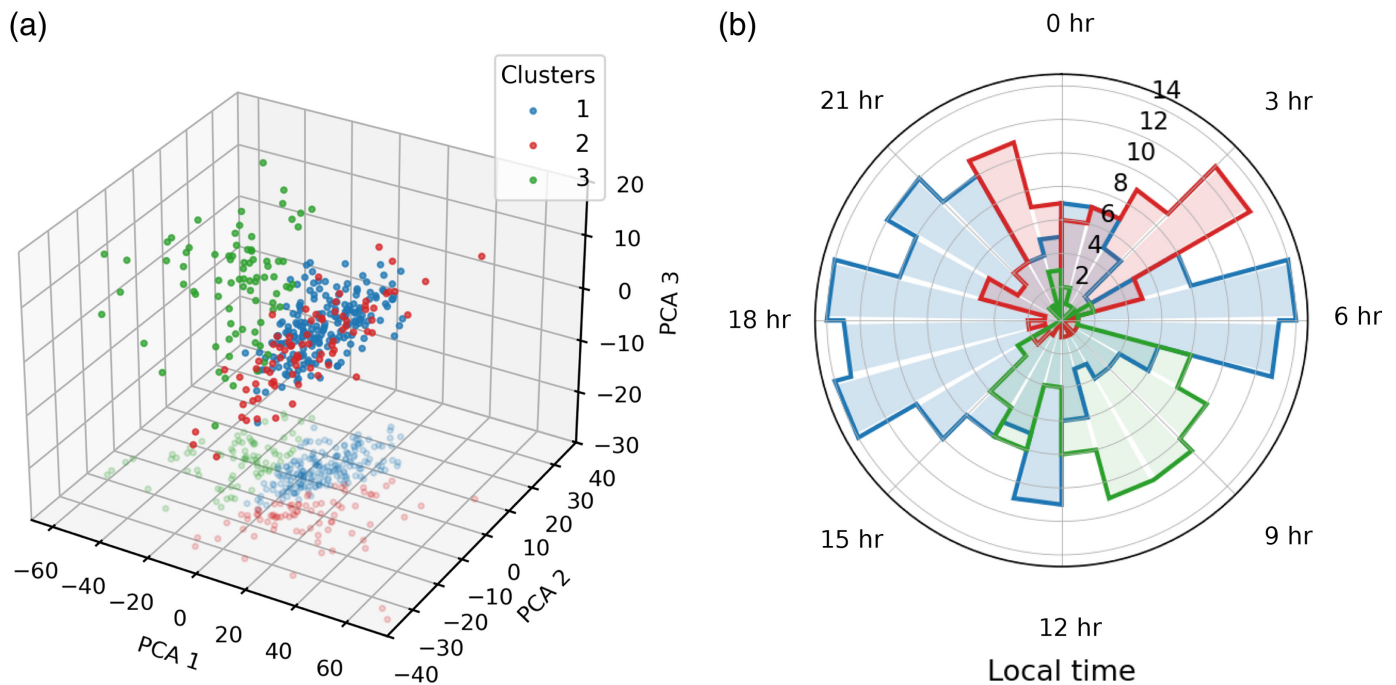
Figure 5. Principal component analysis (PCA) of the representative vectors of the synthetic VSGs. (a) Cumulative variance as a function of the number of PCA components. The red dotted line indicates the advised threshold for capturing the most important features of the data, and the blue dotted line represents our obtained value, which is higher. (b) Projection of the first three PCA components for three clusters in three colors. The color version of this figure is available only in the electronic edition.

Application of clustering on VSGs obtained from DAS ambient noise data

The 360 VSGs obtained from segment A are clustered by applying the same method as for the synthetic dataset. The projection of the first three PCA components, which have the highest variance and carry the most important information about each sample (VSG compressed into PCA components), for the different clusters is shown in Figure 6a. We also present the temporal distribution of the VSG's time of record (hour of the day) within these clusters (Fig. 6b). In general, we observe that our approach has clustered the VSGs according to the three main noise patterns observed during visual inspection and shown in Figure 3.

The first cluster of hourly recordings is distributed throughout the day and night with higher contributions during rush hours (early morning, 04:00–07:00 a.m. and late afternoon to evening, 03:00–08:00 p.m., shown in blue in Fig. 6b). The respective VSGs are higher in quality than those of the other two clusters, that is, with better symmetry of causal and acausal parts and better consistency of dispersion spectra (first row with blue titles in Fig. 7 concerning second and third rows).

The VSG stack corresponding to the second cluster shows strong contamination by monofrequent noise, resulting in more pronounced discontinuities in the corresponding dispersion



spectra (second row with red titles in Fig. 7). The hourly recordings of this cluster predominantly occur during nighttime (08:00 p.m.–04:00 a.m.), as indicated by the red distribution in Figure 6b, suggesting that these signals may originate from periodically operating machinery in the vicinity, such as pumps or power transformers, that are either inactive during the day or their seismic noise drowned in traffic-related signals.

The VSG stack for the third cluster is characterized by the deviation of the surface-wave moveout at an optical distance of ~ 2350 m. In addition, its corresponding dispersion spectrum shows the presence of higher modes, along with a discontinuity in the dispersion curve of the fundamental mode around 1.5 Hz, in which the velocity is higher compared to the dispersion spectra of the first cluster (third row with green titles in Fig. 7). This cluster is most active during standard business hours (07:00 a.m.–03:00 p.m.), as indicated by the green distribution in Figure 6b, suggesting that the source operates primarily during the day. Based on the inspection of the area, we conclude that the deviation of the extracted signal in the VSG is probably the result of a workshop at this optical range of the array.

To compare the dispersion spectra of the three clusters, we picked the dispersion curves using a semiautomatic method. For each frequency of the dispersion spectra, the highest amplitude within a predefined frequency interval near the fundamental mode of the dispersion spectrum was selected (Ehsaninezhad *et al.*, 2024). The resulting picked dispersion curves are shown in Figure S5. In contrast to the fundamental mode of the first cluster, the fundamental mode of the second and third is less smooth in the relevant frequency band between ~ 1 and 4 Hz. At frequencies below 1.5 Hz, the second and third clusters exhibit strong fluctuations and an abrupt steep ascent in the lower frequency range (<1.5 Hz). Further fluctuations can be observed in the

Figure 6. (a) Projection of the first three PCA components for three clusters in three colors (for better illustration of different clusters, the projection of PCA components onto the first two components is also shown on the floor of the 3D plot with less intense colors). (b) The recording time of the VSGs is distributed in each respective cluster across the 24 hours of the day. The color version of this figure is available only in the electronic edition.

frequency range above 3 Hz, in which a higher mode is more pronounced in the dispersion spectra, and the automatic picking selects the wrong local maximum as a consequence. The influence of the contaminated VSGs in clusters 2 and 3 is even evident when considering the dispersion spectra obtained from stacking all 360 VSGs (orange curve in Fig. S5), that is, without our clustering procedure. An increase in the velocity in the 1–1.5 Hz frequency range is still visible. Given these observations, we selected the first cluster for further analysis.

We repeated the analysis for the three other cable segments B, C, and D. The dispersion spectra from stacked VSGs, clustered using our method, show similar qualitative differences as in segment A, allowing clear visual distinction between clusters (Figs. S11, S13, and S15). However, unlike segment A, segments B, C, and D each exhibit only two main VSG clusters. The timing of these VSG recordings correlates with day and night cycles: the lower-quality VSGs are typically recorded at night and display signs of monofrequent noise contamination (Figs. S10, S12, and S14).

The absence of a third cluster in these segments is likely due to the lack of a strong, localized noise source along the array. Overall, the analysis across the four separate segments indicates heterogeneous noise conditions, with localized noise

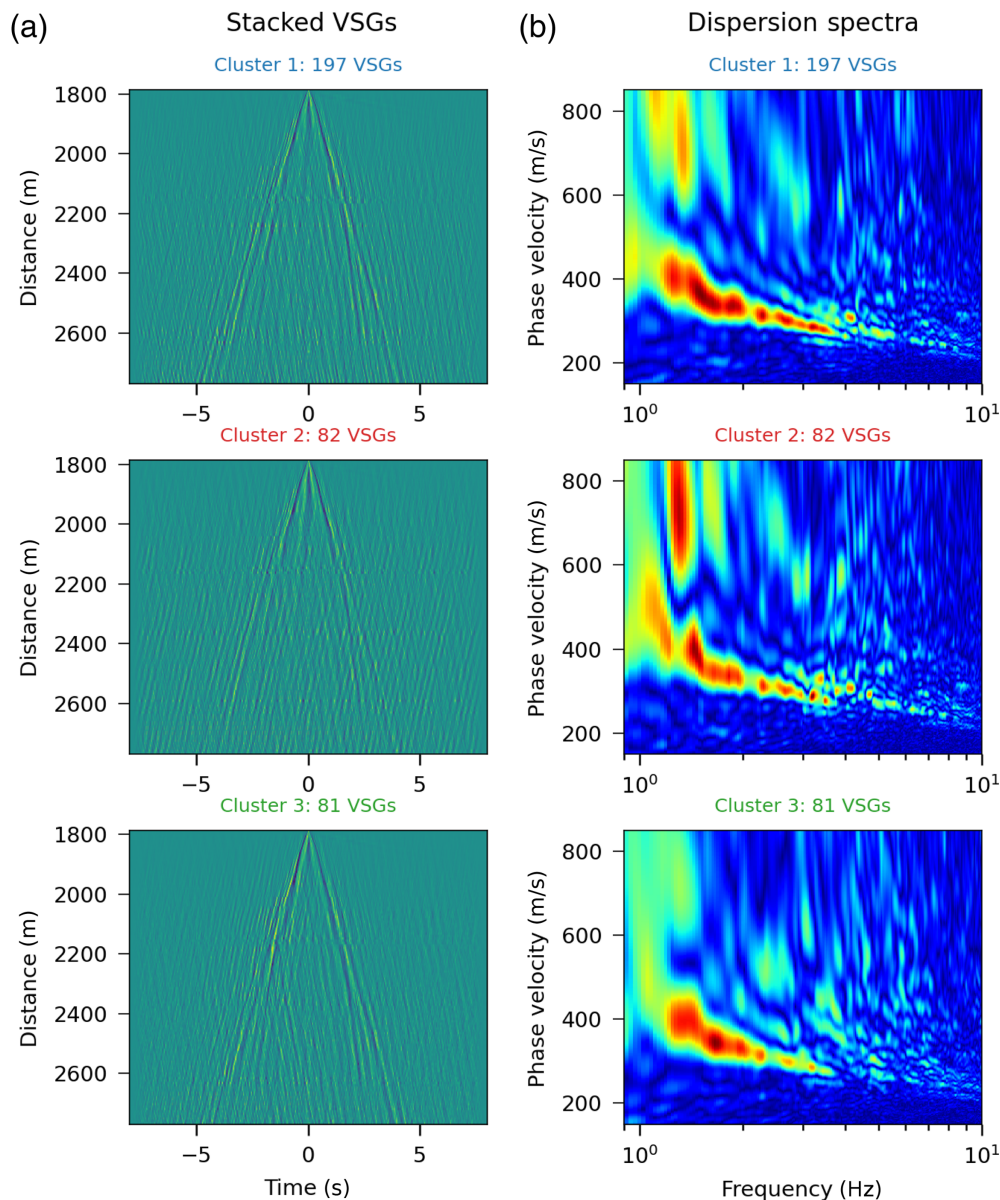


Figure 7. (a) Stack of clustered VSGs for segment A by cluster (rows). (b) The corresponding dispersion spectra. The color version of this figure is available only in the electronic edition.

affecting only certain parts of the array. Although monofrequent noise appears throughout the array, its intensity varies, influencing the number of high-quality VSGs. For example, segment C shows a significantly reduced number of qualitative VSGs compared to segments B and D (Figs. S11–S15).

1D shear-wave velocity profile

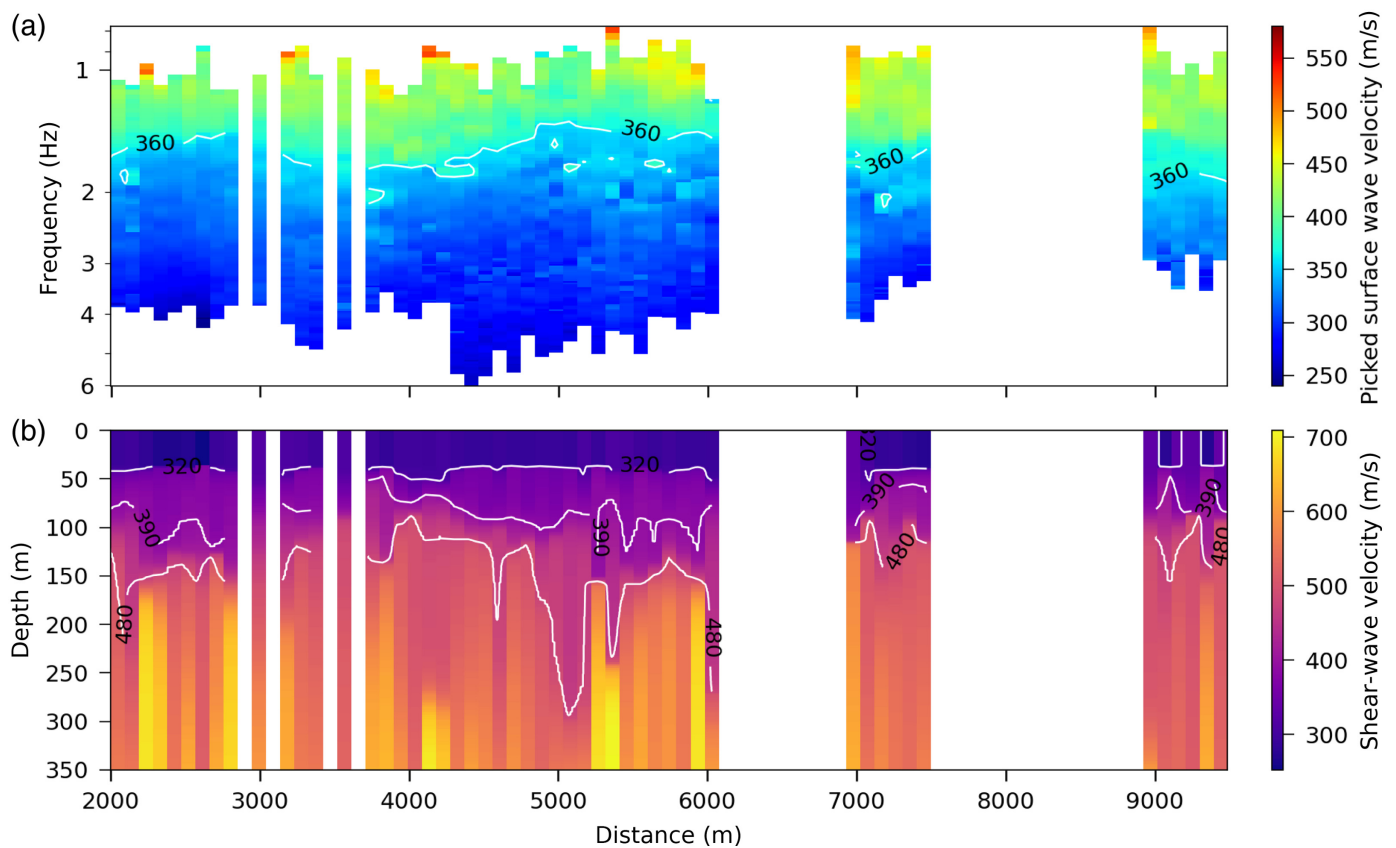
After the clustering of all calculated VSGs, we evaluate their dispersion spectra and select the best cluster with the highest quality VSGs. For example, in segment A, cluster 1 in Figure 7 was chosen for the next steps. The picked fundamental mode of the chosen dispersion spectra is then inverted to obtain the 1D shear-wave velocity underneath this segment. The inversion is

performed using a stochastic direct-search method (neighborhood algorithm) with a model that consists of four layers on top of the half-space, each with predefined upper and lower velocity bounds (Sambridge, 1999; Ehsaninezhad *et al.*, 2024). The 1000 best-fitting shear-wave velocity models obtained for segment A are shown in Figure S16b, together with the median velocity model, which is derived by resampling these 1000 models at 1 m depth intervals and calculating the median velocity for each resampled layer. The 1D model agrees very well with the model obtained by visual selection of VSGs in Ehsaninezhad *et al.* (2024) (Fig. S16a). We refer the readers to Ehsaninezhad *et al.* (2024) for more details about the inversion process, parameters, and validation.

2D velocity model

We adopt the same workflow (VSGs extraction, clustering, and manual selection of the best cluster for MASW) for all the overlapping subsegments into which we divided the straight fiber section. Segments that do not yield a proper virtual shot gather, possibly due to either the presence of persistent local noise (e.g., Fig. S8) or bad coupling conditions of the cable (e.g., Fig. S9), are excluded before clustering. As a result, there are observational gaps in the section of the picked dispersion curves shown in Figure 8a, in which the picked phase velocity of the fundamental mode is color-coded as a function of distance and frequency. The frequency range of the picked dispersion curves varies from segment to segment, which impacts the segment-wise shear-wave velocity inversions.

The final pseudo-2D velocity model (Fig. 8b) is constructed by concatenating the 1D median velocity profiles of each fiber segment (also see Fig. S16). The variation and maximum extent of the distribution of the model interfaces are shown in Figure S18 using box-and-whiskers plots. The 2D shear-wave velocity model shows a relatively simple “layer-cake” structure. We



identify a first layer extending from the surface to the 320 m/s contour line, which lies at depths ranging between 40 and 60 m. The 390 m/s contour line in Figure 8b may delineate the base of a second layer that spans from 50 to 120 m depth, showing a gradual velocity increase from 320 to 390 m/s. It is interesting to note that the thickness of this layer behaves in opposition to the 360 m/s contour line of the picked phase velocity in Figure 8a, for example, a phase velocity of ~ 360 m/s at frequencies of ~ 1.5 Hz at 2800 and 5000 m distance correlates with a deepening of the 390 m/s shear-wave velocity contour in the 2D velocity model. In contrast, at a 3200 m distance, where the picked phase velocity in Figure 8a reaches a frequency of 1.8 Hz, the second layer is narrower in Figure 8b.

We interpret a third layer that is bound by the 390 and 480 m/s shear-wave velocity contours, and a fourth and last layer extending from the 480 m/s contour line to the bottom of the profile. At these greater depths, the lateral variations of the shear-wave velocity contours in Figure 8b become more pronounced, which agrees with the irregular picking of phase velocities at frequencies below 1 Hz (Fig. 8b). A higher standard deviation is observed at greater depths in Figure S17, which indicates that the model's uncertainty increases with depth, making it less reliable at greater depths.

Report on the DAS data usability

After excluding low-quality segments (white gaps as shown in Fig. 8) and the associated hourly recordings, we divided the

Figure 8. 2D shear-wave velocity model derived for an 8 km long segment along the straight line of the fiber shown as a green dashed line in Figure 1. (a) Picked dispersion curves for each 1D segment along the 2D array. (b) Pseudo-2D shear-wave velocity model. The white gaps correspond to segments for which VSGs of enough quality could not be obtained. The color version of this figure is available only in the electronic edition.

remaining VSGs into three recording periods within a 24 hr day based on the patterns observed in the clustered VSGs. Figure 9a illustrates the percentage of data used to create the final 2D velocity model across the 2D array. Overall, only around 37% of the available VSGs are finally inverted. Notably and in contrast to other ambient noise tomography studies (Guan *et al.*, 2021; Morton *et al.*, 2021; Gisselbrecht *et al.*, 2023), which prioritize nighttime data, only a small fraction of nighttime data was used (6% concerning all data, less than 18% concerning data stemming from the nighttime, and 16% concerning all accepted data). In our study site, the discarded data from the nighttime mostly consists of VSGs contaminated with monofrequent signals. In contrast, data collected during the other hours, mostly daytime, were more usable. The percentage of used data from the rush and business hours is quite similar and significantly outnumbers the proportion of data stemming from the night (Fig. 9b). For our study site, daytime and rush hour data are preferred over nighttime data. However, it is important to note that this preference is specific

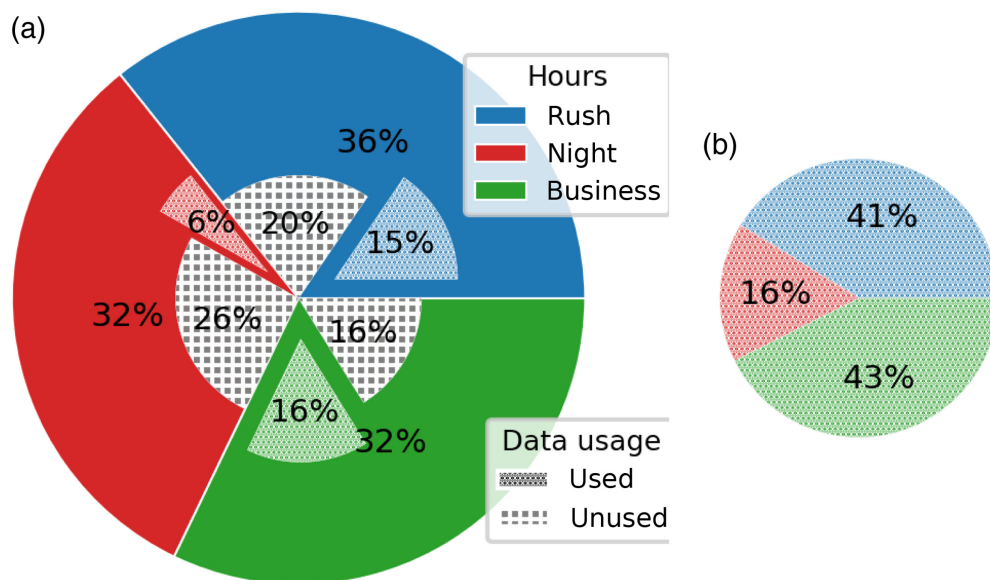


Figure 9. Statistics of recording periods: (a) Data categorization by three different periods of the day, that is, rush (04:00–07:00 a.m. and 03:00–08:00 p.m.), night (08:00 p.m.–04:00 a.m.), and business hours (07:00 a.m.–03:00 p.m.) is represented in the outer pie chart. Within each of these categories, the inner chart differentiates between used and rejected data. (b) Proportion of used data concerning the total final utilized data. The color version of this figure is available only in the electronic edition.

to our test site. Performing this type of analysis is valuable for understanding the unique noise characteristics of the location.

Discussion

The results presented in this study demonstrate that applying ambient noise interferometry to DAS data recorded on telecommunication fibers in urban environments is a promising approach for efficient, high-resolution subsurface imaging. However, it also presents distinct challenges that mainly stem from the complexity of the urban seismic noise. Consequently, precise and rigorous data quality control is essential to ensure the reliability of interferometric results and their interpretations. In our previous study, we demonstrated that denoising the stacked VSGs can improve the results (Ehsaninezhad *et al.*, 2024). However, this is only true when the number of low-quality VSGs included in the stack is not excessively high. Although quality control of VSGs has been done manually in the past, we now employ unsupervised clustering to achieve the same outcome.

With 15 days of ambient noise data available for subsurface mapping, we first cluster the calculated VSGs according to their similarity. This unsupervised clustering approach works without relying on a reference VSG (e.g., average of all VSGs). Only the cluster containing the high-quality VSGs is kept for further processing. This method is particularly useful when dealing with noisy, high-frequency ambient noise data, such as that recorded in urban environments, in which noise patterns are governed by the diurnal cycle of anthropogenic activity. Furthermore, the DWT coefficients, representing each

VSG, provide better resolution and capture higher variance within the data compared to a workflow in which the characteristic vector (the sample of the PCA) is directly calculated from the VSG.

Despite this improved representation, it is worth mentioning that, when applying PCA, we have to retain a larger number of components (100 PCA components for each VSG) compared to studies dealing with NCFs derived from lower frequency ambient noise with a stronger signal, such as in Viens and Iwata (2020). The reason for this could be the fact that, when using 2D arrays, spatiotemporal areas within the VSG that are off the main surface-wave moveout path play a critical role in categorizing the differ-

ent types of VSG, and therefore, the PCA components associated with these areas should also be retained.

After developing the foundation of the approach using synthetic VSGs, we optimize the workflow for field DAS ambient noise data by making several adjustments to maximize the utility of the data. These adjustments included determining the appropriate number of clusters for VSG categorization based on visual inspection (Ehsaninezhad *et al.*, 2024), which identified three or two main categories of noise, depending on the array section considered. Consequently, we fixed the number of clusters to three or two for each segment.

Although the clustering method showed strong performance on synthetic data, its application to real DAS recordings achieved a lower accuracy. This could be due to the high level of noise in the VSGs. It is also worth noting that our evaluation of the clustering results and consequent selection of the “best” stack amenable to further processing is done manually. We visually inspect the outcome of each cluster (both stacked VSG and dispersion spectra), as identifying a specific parameter to evaluate the results associated with each cluster proved challenging. Developing a more robust criterion for evaluating the results of each cluster will pave the way for a fully automated workflow in the future.

To further evaluate the performance of the clustering approach and provide a more quantitative analysis, we compare our clustering results with the manual selection of VSGs conducted in Ehsaninezhad *et al.* (2024) by calculating a confusion matrix of the two sets of labels (Fig. 10). Diagonal elements

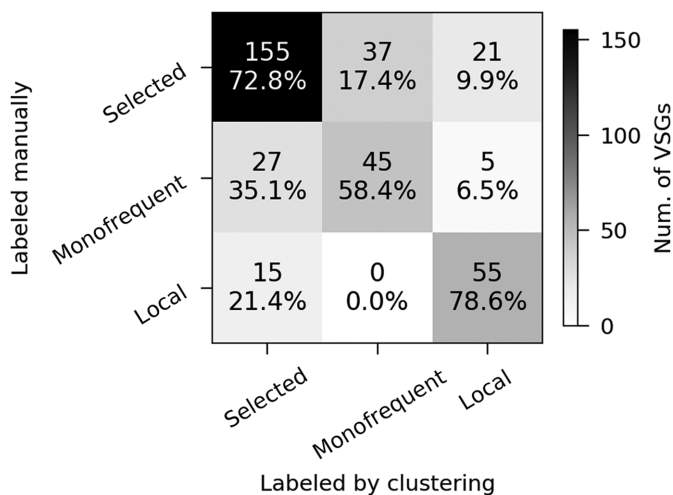


Figure 10. Confusion matrix comparing the categorization of VSGs from fiber segment A into three clusters. Categorization was achieved through the clustering method and manual inspection. The main diagonal shows the number of VSGs assigned to the same category by both methods. The percentages represent the proportion of VSGs assigned to a category through clustering, relative to the total number of VSGs that were manually categorized.

represent the number of VSGs correctly assigned to their respective clusters. Off-diagonal elements indicate miscategorized VSGs across clusters. The percentages shown in each element of the matrix correspond to the proportion of the clustered VSGs concerning the total number of VSGs manually associated to a category (e.g., of all VSGs “selected” during the manual labeling, 72.8%, 17.4%, and 9.9% were labeled “selected”, “monofrequent (noise)”, and “local (noise source)”, respectively). The total accuracy of the clustering is calculated by dividing the sum of the diagonal elements by the total sum of all values in the confusion matrix. The clustering results achieve an overall accuracy of ~71%. The biggest fraction of miscategorization occurs in cluster 2 (“monofrequent noise”), in which 37 VSGs are assigned to cluster 1 (“selected”). However, given the long duration of the campaign, this limited degree of miscategorization of VSGs did not significantly affect the final results. For shorter-duration campaigns with limited data, however, we recommend implementing more stringent quality control measures such as shorter time data segmentation and noise-source separation because contamination of VSGs could affect the outcomes, as emphasized by Ehsaninezhad *et al.* (2024).

The differences in the final dispersion spectra can depend on the segment's exposure to different noise sources and the proportion of low-quality VSGs (see Fig. S7). For example, in segments A, B, and D, the number of low-quality VSGs is relatively low, and so is their impact on the final dispersion spectra. In contrast, in segment C, in which a larger proportion of low-quality VSGs is present, we observe a significant reduction in the quality of the resulting dispersion spectra (Fig. S7).

In terms of computational cost, the most demanding steps are the standard preprocessing of DAS ambient noise recordings and the calculation of cross correlations over 360 hr of data. In comparison, the computation of DWT coefficients, PCA, and clustering is relatively lightweight (e.g., in a conventional laptop, it takes only a few minutes). This is because preprocessing and cross-correlation are applied to full 1 hr recordings, whereas each VSG consists of only 14 s of down-sampled signal. Therefore, incorporating a quality control step in the evaluation of clustering results is beneficial. It helps ensure that VSGs accurately representing structural features are used in further analysis.

Our developed approach also allows us to assess the characteristics of ambient seismic noise, link it to human activities, and determine the times of day that yield the most reliable data. Other studies have also shown the importance of analyzing the noise associated with day or nighttime of the dataset Lai *et al.* (2024). Quality control of the obtained VSGs suggests that nighttime recordings are less favorable at our study site. We hypothesize that monofrequent noise originating in the vicinity of the cable becomes more pronounced at night due to lower overall noise levels and induced vibrations by ongoing traffic. However, in conventional ambient noise tomography studies, it is typically recommended to prioritize nighttime recordings to minimize the impact of strong, localized noise sources associated with daytime human activities (Guan *et al.*, 2021; Morton *et al.*, 2021; Gisselbrecht *et al.*, 2023). These time recordings likely target natural noise or specific traffic noise sources rather than the broader spectrum of high-frequency cultural noise.

In terms of geological interpretation, we demonstrate that the velocity model obtained here using a semiautomatic approach to select valuable data periods is in good agreement with a previously derived 1D velocity model based on manual data selection (Ehsaninezhad *et al.*, 2024), which was shown to agree with regional geological information. Following Ehsaninezhad *et al.* (2024), we interpret the third interface of the here presented pseudo-2D velocity model, which approximately follows the 390 m/s contour of the median velocity model, as the top of the Rupelian layer. This interface varies in depth between 90 and 160 m S18, which is in agreement with stratigraphic studies of the area. The regional model by Schilling *et al.* (2018) indicates the top of this layer at a depth of 140 m. The nearest stratigraphic sample from a research well is at a distance of ~2 km from our array and shows the Rupelian at a depth of 100 m Pei *et al.* (2024). Therefore, the derived model for the top of the Rupelian remains within the boundaries observed in the named stratigraphic studies.

Regional geological information by Schilling *et al.* (2018) indicates a relatively simple layered structure for this area. Moreover, the alignment of lateral phase velocity fluctuations at a range of over 1 km (Fig. 8a) with lateral variations of the interface depths of the upper three layers (Fig. 8b) suggests that

the inverted shear-wave velocity model is well constrained by the data at this depth range, and it likely represents real geological features. However, in the deepest layer, abrupt and unrealistic velocity contrasts appear, which correspond to segments with high picked phase velocities at low frequencies (<1 Hz; Fig. 8a). This indicates that low-frequency data are poorly constrained, and the large lateral variations observed at depths from 200 m downward are likely artifacts of the inversion process.

Conclusion

We present a case study that leverages DAS deployed on existing telecommunication networks and urban ambient noise for seismic imaging of the subsurface. Our approach involves enhanced quality control through unsupervised clustering of VSGs extracted from 15 days of continuous DAS ambient noise data. The clustering results on the synthetic dataset demonstrate full accuracy, validating the effectiveness of this method. In field data, the clustering (or categorization) of VSGs derived from DAS urban ambient noise facilitates the removal of low-quality data, consequently enhancing the accuracy of dispersion spectra and ultimately leading to more precise estimations of subsurface shear-wave velocities. Although our semiautomated approach allows for a more efficient management of the data and shows improvements over standard DAS ambient noise data processing, we acknowledge the need for manual inspection of clustering results.

In addition, this categorization helps identify distinct DAS ambient noise patterns across the city. Our findings indicate that the recorded noise on dark fiber in urban areas exhibits strong time dependency. This variability is mostly influenced by the noise sources. However, with a clear understanding of which time slots are optimal, such as during the late afternoon to evening hours, we can use shorter DAS data recording periods to achieve higher quality results. Our workflow is broadly applicable to other urban DAS datasets, although it might require some degree of site-specific customization, such as selecting the optimal number of clusters or defining the number of PCA components to suit different urban environments. Further improvements may be achieved by enhancing the classification or clustering of DAS noise recordings and developing predictive models for urban DAS ambient noise.

Data and Resources

Python was used for data analysis and plotting (Hunter, 2007; Beyreuther et al., 2010). In particular, we used scikit learn Python package to perform clustering and principal component analysis (PCA) (Pedregosa et al., 2011) and PyWavelets Python package for wavelet coefficient calculation (Lee et al., 2019). Geopsy package 3.4.2 (<https://www.geopsy.org>, last accessed September 2023) was used for the inversion process. The map and the insets in Figure 1 are based on data from (OpenStreetMap contributors, 2017) and were downloaded with contextily-python API (<https://contextily.readthedocs.io>, last accessed September 2023) in which the insets use map tiles by Stamen Design,

under CC BY 4.0. All subsets of hourly decimated strain-rate recordings associated with segment A shown in Figure 1 and a list of channels within loops of slack are available online (at doi: [10.5880/GFZ.2.2.2023.003](https://doi.org/10.5880/GFZ.2.2.2023.003)). The supplemental material includes synthetic results, as well as results for segments B, C, and D, a comparison of cumulative variance for waveform, power spectral density (PSD), and wavelet coefficients, examples of excluded segments, along with the standard deviation and depth distribution of the pseudo-2D velocity model.

Declaration of Competing Interests

The authors acknowledge that there are no conflicts of interest recorded.

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